

Quantum magic and Conformal Field Theory

Masahiro Hoshino
Department of Physics, UTokyo

Masaki Oshikawa
ISSP, UTokyo

Yuto Ashida
Dept. Phys. & IPI, UTokyo



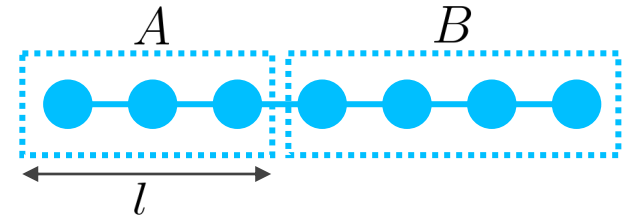
Scaling of Entanglement Entropy in 1D pure states

Gapped ground states

M B Hastings, J. Stat. Mech (2007)

◆ efficient simulation with Matrix Product States (MPS)

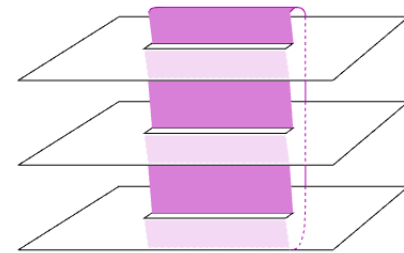
◆ area law $S_A \sim \text{const.}$



Critical (gapless) ground states

◆ universal scaling determined by the central charge

◆ log law $S_A \sim \frac{c}{3} \ln l$

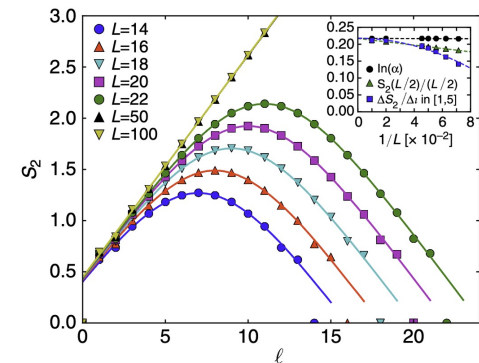


P. Calabrese & J. Cardy, J. Stat. Mech. (2004,2009)

Thermal (typical) pure states

◆ almost maximally entangled states

◆ volume law $S_A \sim l$



Y. O. Nakagawa, et al., Nat. Commun. (2018)

EE reflects phases of matter!

Entanglement: resource for quantum communication

Local operations & classical communication (LOCC)

$$|0\rangle|0\rangle \xrightarrow{\text{X}} \frac{|0\rangle|0\rangle + |1\rangle|1\rangle}{\sqrt{2}}$$

cannot create entanglement

Separable state

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

LOCC

≈ quantum channel acting locally

$$\mathcal{E} = \sum_j \mathcal{E}_j^A \otimes \mathcal{E}_j^B$$

Entangled state

$$\rho \neq \sum_i p_i \rho_i^A \otimes \rho_i^B$$

Entangling gates & measurements

ex)

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad |\text{Bell}\rangle\langle\text{Bell}|$$

Entanglement entropy (EE) : entanglement measure for pure states

$$S_A = -\text{Tr}[\rho_A \ln \rho_A]$$

$|\psi\rangle$: pure state

$\rho_A = \text{Tr}_B[|\psi\rangle\langle\psi|]$: reduced density matrix

General framework: quantum resource theory

Resource $\approx$ things that cannot be **easily** obtained

example: entanglement, nonstabilizerness (magic), non-Gaussianity, quantum coherence, ...

Free states

restricted class of states (easily prepared)

Free operations

operations acting invariantly on free states

easy

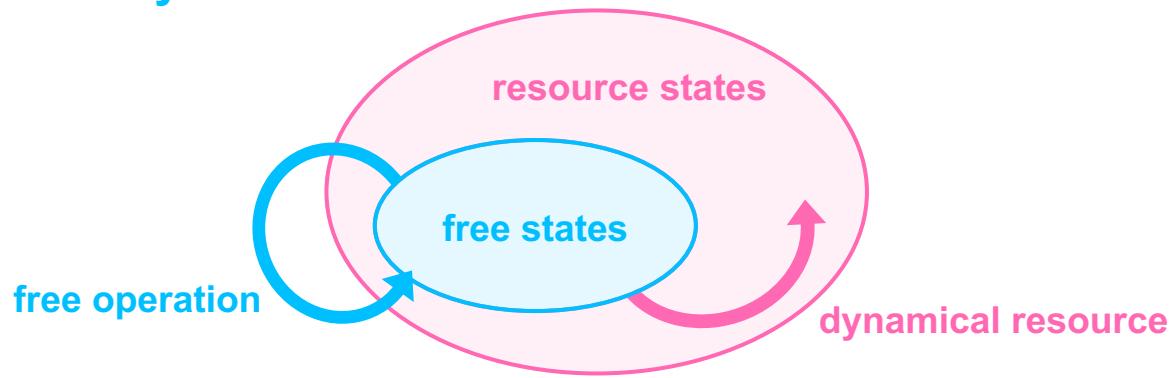
Resource states

states that are not free

Dynamical resources

operations that are not free

hard



Nonstabilizerness: resource for universal quantum computation

Universal gate set = **Clifford** + **T**

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Clifford unitary

- easy to implement fault-tolerantly (transversal gates)
- classically simulable due to Gottesman-Knill theorem

T gate

Stabilizer states

(convex hull of) **Clifford** $|0\rangle^{\otimes L}$

Stabilizer protocol

Clifford + other easy operations

Non-stabilizer state (magic state)

$\neq$ **Clifford** $|0\rangle^{\otimes L}$

Non-Clifford gates

ex) T gate, CCZ gate

cf. Eastin-Knill theorem

Nonstabilizerness: resource for universal quantum computation

Universal gate set = **Clifford** + **T**

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Clifford unitary

- easy to implement fault-tolerantly (transversal gates)
- classically simulable due to Gottesman-Knill theorem

T gate

Example: Bell state $|\text{Bell}^{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

◆ entanglement: **resource state**

$$S_A = \ln 2 > 0$$

◆ nonstabilizerness: **free state**

$$|\text{Bell}^{00}\rangle = \text{CNOT } H |00\rangle$$

$$H |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\begin{aligned} \text{CNOT}(a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle) \\ = a|00\rangle + b|01\rangle + c|11\rangle + d|10\rangle \end{aligned}$$

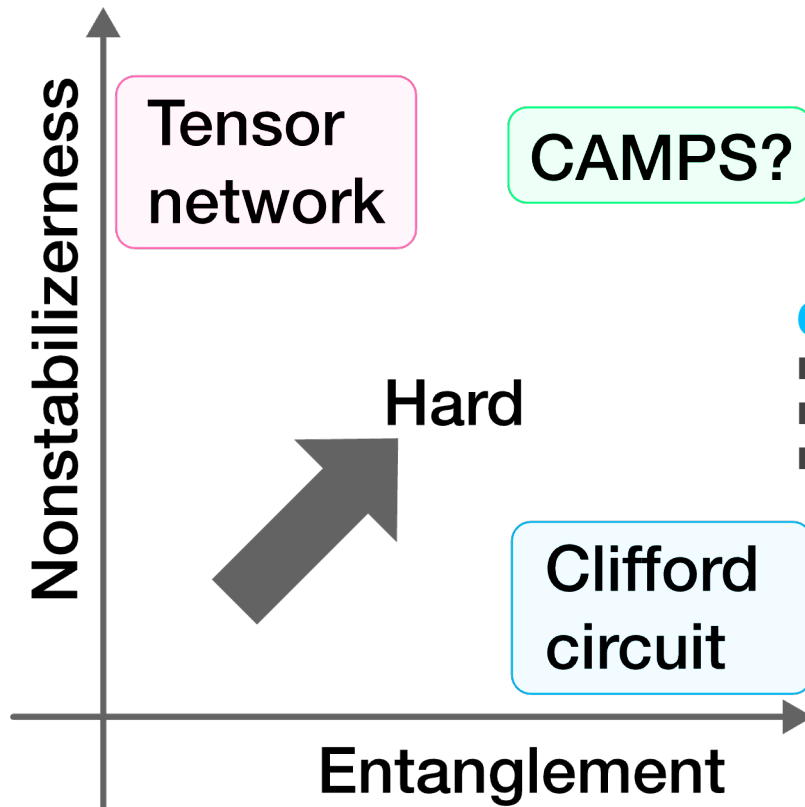
Even highly entangled states can be simulated classically!

Hardness of classical simulation

Entanglement alone does not indicate “quantumness”

Tensor network

- able to simulate low-entangled states
- cost $\approx$ bond dimension χ
- scales as $\chi \propto \exp(S_A)$



Clifford circuit

- able to simulate low-magic states
- cost $\approx$ number of T-gates
- scales as time $\sim \text{poly}(n, m) + \exp(T)$
(n: # of qubits, m: # of Clifford gates, T: # of T-gates)

Bravyi & Gosset, PRL (2016) / arXiv:1601.07601

Quantifying magic

Amount of magic $\approx$ distance from stabilizer states

Stabilizer Rényi entropy (SRE): measure for pure states

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \sum_{\vec{m}} \frac{\text{Tr}^{2\alpha}[\sigma^{\vec{m}}\psi]}{2^L}$$

$\psi = |\psi\rangle\langle\psi|$: L -qubit pure state

$\sigma^{\vec{m}} = \sigma^{m_1} \sigma^{m_2} \dots \sigma^{m_L}$: Pauli string

Properties

L. Leone et al., PRL (2022)

- Faithfulness** : $M_\alpha(\psi) = 0 \iff$ stabilizer state $\because \text{Tr}[\sigma^{\vec{m}}\psi] = 1$ if $\sigma^{\vec{m}} \in S, |S| = 2^L$
- Additivity** : $M_\alpha(\psi \otimes \phi) = M_\alpha(\psi) + M_\alpha(\phi)$
- Monotonicity** : non-increasing under stabilizer protocol
only for $\alpha \geq 2$

Operational meaning

see also L. Bittel & L. Leone, arXiv:2507.22883

Stabilizer protocol : $|R_1\rangle^{\otimes M} \rightarrow |R_2\rangle^{\otimes N}$

$$\frac{N}{M} \leq \frac{M_{\alpha \geq 2}(R_1)}{M_{\alpha \geq 2}(R_2)}$$

conversion process

rate of conversion $\leq$ ratio of SRE

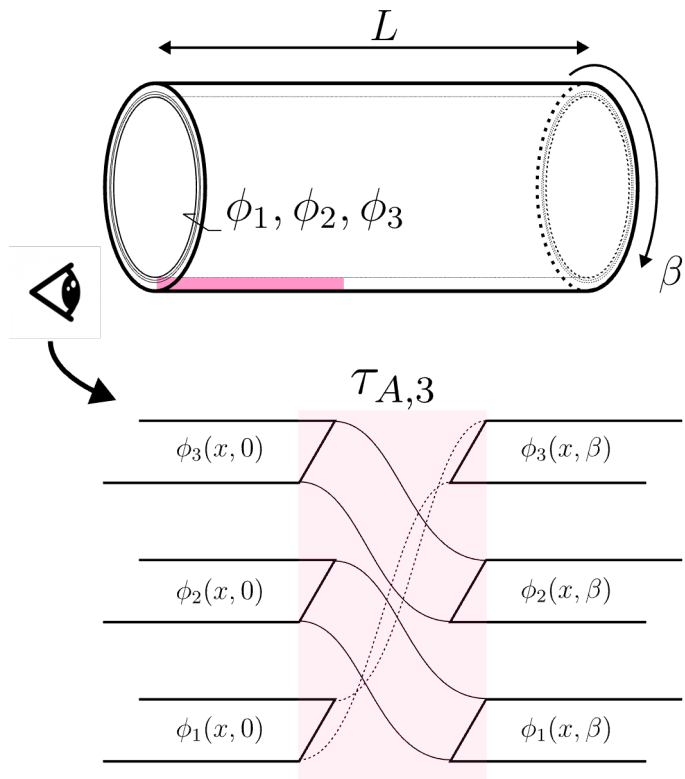
Entanglement entropy vs Stabilizer entropy

$$S_A^{(\alpha)} = \frac{1}{1-\alpha} \ln \text{Tr}[\rho_A^\alpha] \quad (\rho_A = \text{Tr}_{\bar{A}} \psi)$$

$$= \frac{1}{1-\alpha} \ln \text{Tr}[\tau_{A,\alpha} \psi^{\otimes \alpha}]$$

Permutation operator
invariant under arbitrary unitaries

$$Z_3 = \text{Tr}[\tau_{A,3}(e^{-\beta H})^{\otimes 3}]$$

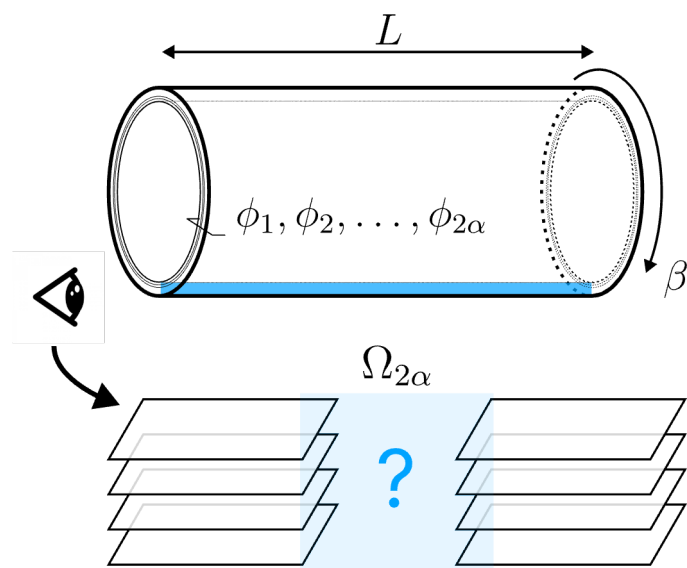


$$M_\alpha = \frac{1}{1-\alpha} \ln \text{Tr}[\Omega_{2\alpha} \psi^{\otimes 2\alpha}]$$

$$\left(\Omega_{2\alpha} = \frac{1}{2^L} \sum_{\vec{m}} (\sigma^{\vec{m}})^{\otimes 2\alpha} \right)$$

Clifford commutant
invariant under arbitrary Clifford unitaries

$$Z_{2\alpha} = \text{Tr}[\Omega_{2\alpha}(e^{-\beta H})^{\otimes 2\alpha}]$$



**Calculation reduces to
determining the boundary condition!**

SRE as participation entropy in the Bell basis

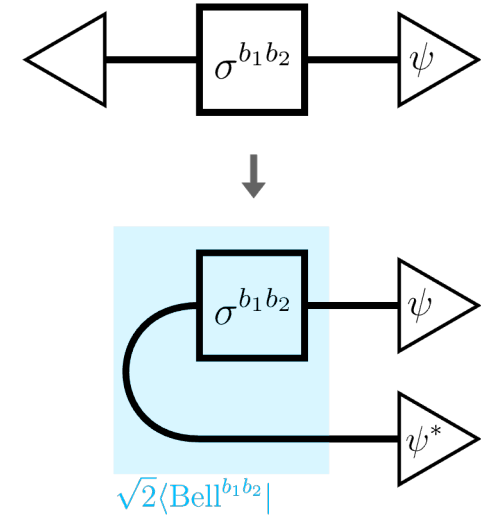
Choi-Jamiołkowski isomorphism

$$\text{Tr}[(\sigma^{\vec{m}})^{\otimes 2} \psi^{\otimes 2}] = 2^L \text{Tr}[P^{\vec{m}}(\psi \otimes \psi^*)]$$

Participation entropy

$$\text{PE} = \frac{1}{1-\alpha} \ln \sum_i p_i^\alpha \quad (p_i = |\langle i | \Psi \rangle|^2)$$

- classical entropy in an orthonormal basis
- Well-explored for 1D CFT in computational basis



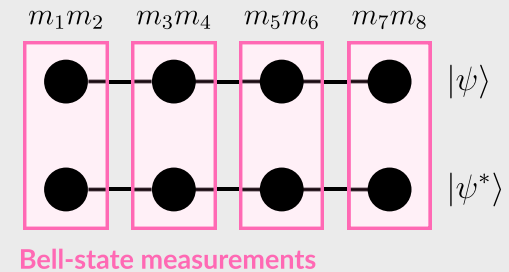
Stephan et al., PRB (2009)
Ashida, Furukawa, Oshikawa, PRB (2024)

...

SRE = participation entropy in the Bell basis

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \sum_{\vec{m}} \underbrace{\text{Tr}^\alpha[P^{\vec{m}}(\psi \otimes \psi^*)]}_{p_{\vec{m}}^\alpha} - (\ln 2)L$$

- constant offset to ensure faithfulness
- PE with $|i\rangle = \bigotimes_{j=1}^L |\text{Bell}^{m_{2j-1}m_{2j}}\rangle$ and $|\Psi\rangle = |\psi\rangle|\psi^*\rangle$



see also: Haug and Kim, PRXQ (2023)

Replica trick + Folding trick → Boundary CFT

Replica trick

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \sum_{\vec{m}} \text{Tr}^\alpha [P^{\vec{m}}(\psi \otimes \psi^*)] - (\ln 2)L$$

$$= \frac{1}{1-\alpha} \ln \text{Tr} \left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes \alpha} (\psi \otimes \psi^*)^{\otimes \alpha} \right] - (\ln 2)L$$

projection operator

partition function of 2α -component CFT with a line defect

Folding trick

$$\frac{Z_{2\alpha}}{Z^{2\alpha}} = \text{Tr} \left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes \alpha} (\psi \otimes \psi^*)^{\otimes \alpha} \right]$$

torus of size $L \times \beta$



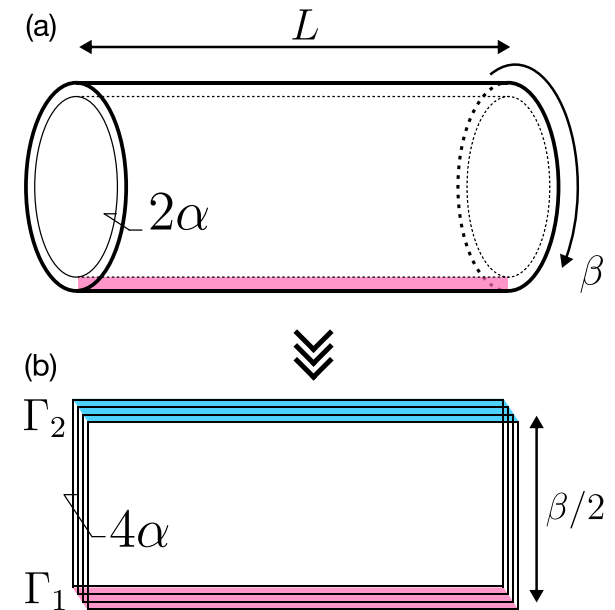
$$Z_{2\alpha} = \langle \Gamma_2 | e^{-\frac{\beta}{2} H} | \Gamma_1 \rangle$$

cylinder of circumference L , length $\beta/2$

H : Hamiltonian of 4α -component CFT on a cylinder

boundary state originating from the line defect

boundary state of the artificially created boundary



Result: universal constant term determined by the g-factor

Noninteger ground-state degeneracy (g-factor)

$$\ln \frac{Z_{2\alpha}}{Z^{2\alpha}} = bL + \ln g + o(1)$$

1. non-universal line energy
2. universal constant term = g-factor $g = g_1 g_2$

$$g_1 = \langle \text{GS} | \Gamma_1 \rangle \quad \text{determined by the boundary condition}$$

$$g_2 = \langle \text{GS} | \Gamma_2 \rangle \quad \text{trivial}$$

$$Z_{2\alpha} \simeq \langle \Gamma_1 | \text{GS} \rangle \langle \text{GS} | \Gamma_2 \rangle e^{-\frac{\beta}{2} E_{\text{GS}}(L)} \quad E_{\text{GS}}(L) = -\frac{\pi c}{6L}$$

$(\beta \gg L)$

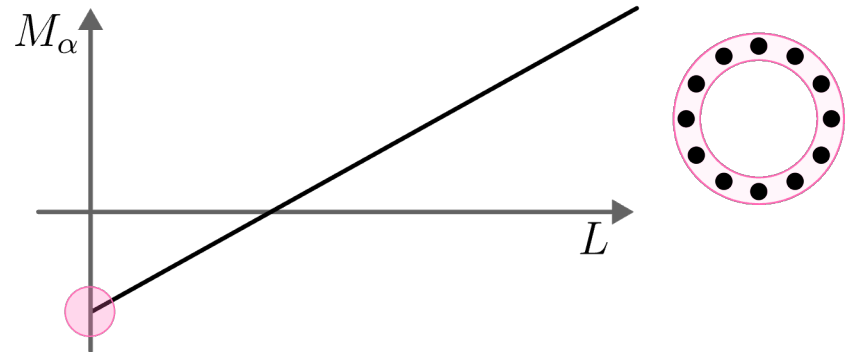


Universality class of the SRE is characterized by the g-factor

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \frac{Z_{2\alpha}}{Z^{2\alpha}} - (\ln 2)L = m_\alpha L - c_\alpha + o(1)$$

$$c_\alpha = \frac{\ln g_1}{\alpha - 1}$$

universal constant term
derived from the g-factor of Γ_1



Example: Ising model

$$H_{\text{Ising}} = - \sum_{j=1}^L (Z_j Z_{j+1} + \lambda X_j)$$

g-factor $g_1 = \langle \text{GS} | \Gamma_1 \rangle = \sqrt{\alpha}$

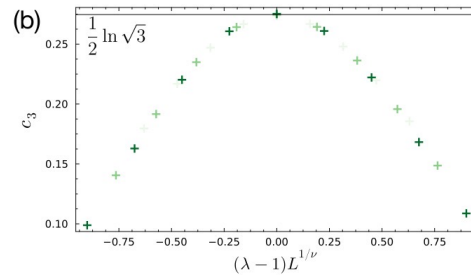
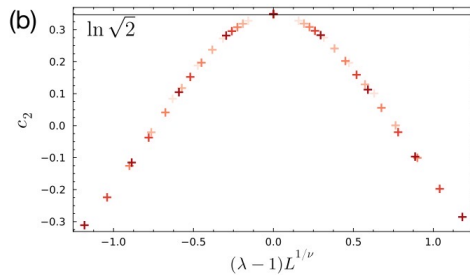
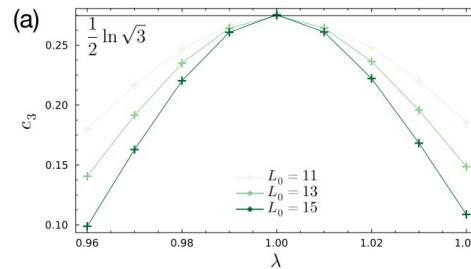
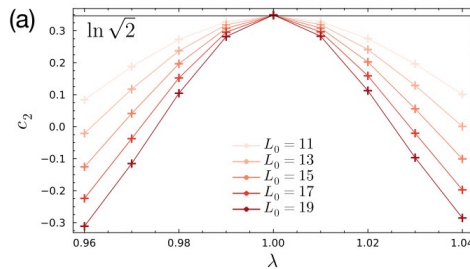
universal constant term

$$c_\alpha = \frac{\ln \sqrt{\alpha}}{\alpha - 1}$$

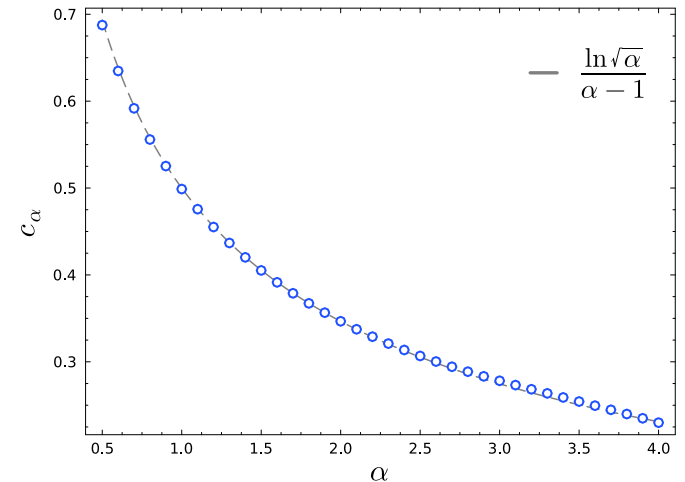
see MH, M. Oshikawa, Y. Ashida, arXiv:2503.13599
for derivation

$$c_2 = \ln \sqrt{2}$$

$$c_3 = \frac{1}{2} \ln \sqrt{3}$$



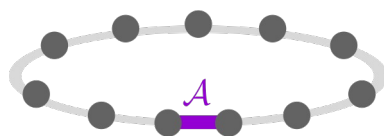
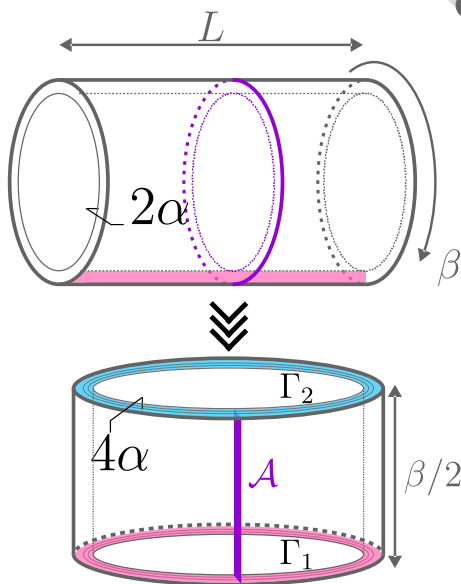
replica Pauli MPS method Tarabunga et al., PRL 133, 010601 (2024)



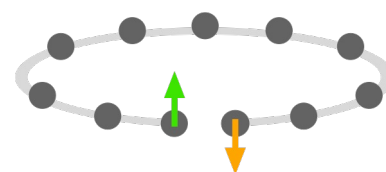
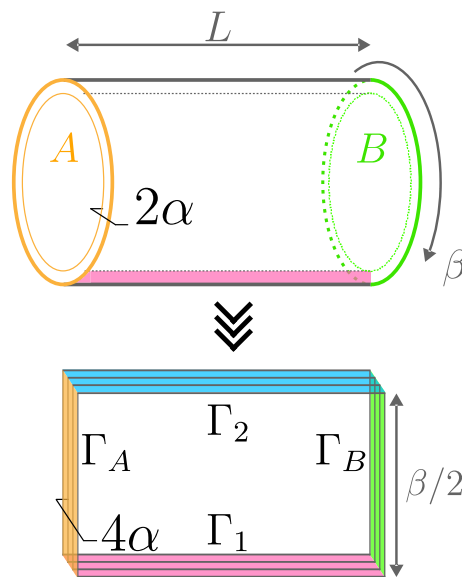
perfect Pauli sampling method G. Lami & M. Collura, PRL 131, 180401 (2023)

Stabilizer Rényi entropy with conformal defects

Topological defect
(totally transmissive)



Open boundaries
(totally reflective, factorising)

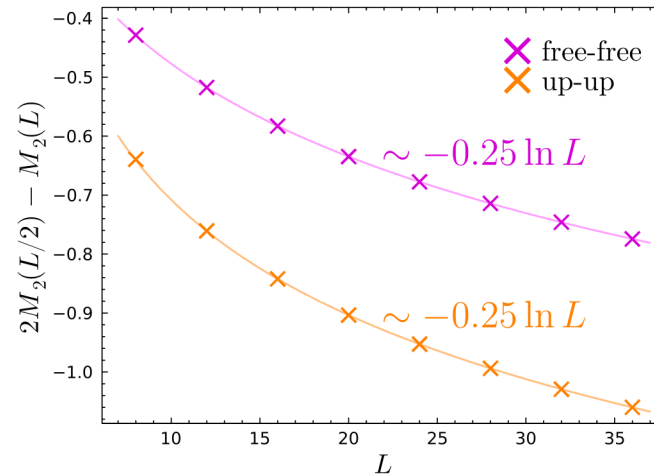
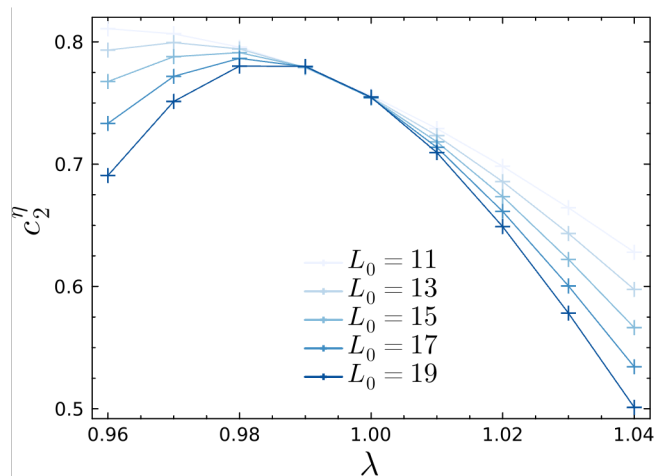


$$M_\alpha = m_\alpha L - c_\alpha^A$$

different bulk sector
different constant term

$$M_\alpha = m_\alpha L + \gamma_\alpha \ln L$$

logarithmic term due to
the corners



Ising CFT: conformal defects move and fuse via Clifford unitaries

Cardy states $|f\rangle, |\uparrow\rangle, |\downarrow\rangle$

Verlinde lines $1, \eta, \mathcal{D}$

N. Seiberg, et al., SciPost (2024)

Fusion rules

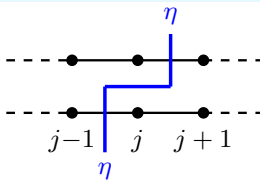
$$\begin{aligned} \eta * |f\rangle &= |f\rangle, & \eta * |\uparrow\rangle &= |\downarrow\rangle, & \eta * |\downarrow\rangle &= |\uparrow\rangle, \\ \mathcal{D} * |f\rangle &= |\uparrow\rangle + |\downarrow\rangle, & \mathcal{D} * |\uparrow\rangle &= |f\rangle, & \mathcal{D} * |\downarrow\rangle &= |f\rangle. \end{aligned}$$

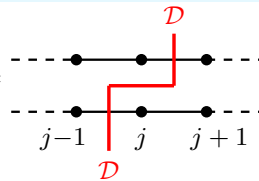
$$\eta \otimes \eta = 1, \quad \eta \otimes \mathcal{D} = \mathcal{D} \otimes \eta = \mathcal{D},$$

$$\mathcal{D} \otimes \mathcal{D} = \mathcal{T}^- \oplus \mathcal{T}^- \eta. \quad \text{Lattice translation defect } \mathcal{T}^-$$

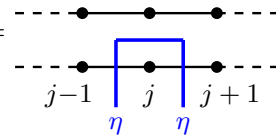
$$\text{In the continuum: } \mathcal{N} \otimes \mathcal{N} = 1 \oplus \eta$$

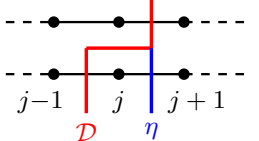
Movement operators

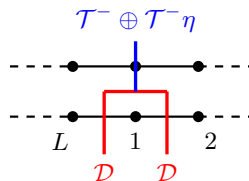
$$U_\eta^j = X_j =$$


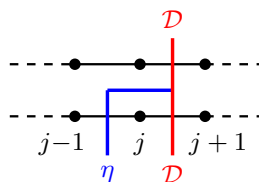
$$U_{\mathcal{D}}^j = \text{CZ}_{j,j+1} H_j =$$


Fusion operators

$$\lambda_{\eta \otimes \eta}^j = X_j =$$


$$\lambda_{\mathcal{D} \otimes \eta}^j = U_{\mathcal{D}}^j X_j =$$


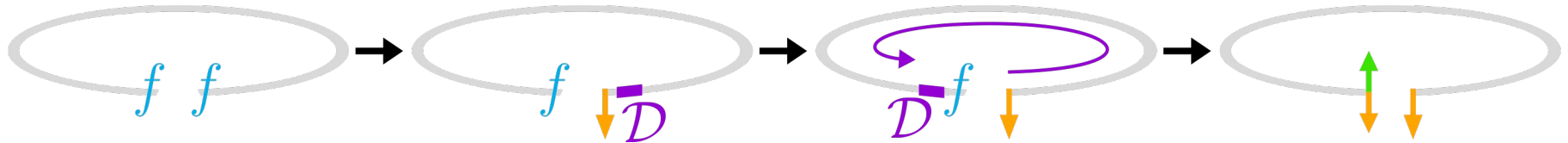
$$\lambda_{\mathcal{D} \otimes \mathcal{D}}^1 = H_1 \text{CZ}_{1,2} =$$


$$\lambda_{\eta \otimes \mathcal{D}}^j = Z_j X_{j+1} =$$


also for fusion with boundaries

SRE encodes fusion rules

$$\mathcal{D} * |\downarrow\rangle = |f\rangle, \quad \mathcal{D} * |f\rangle = |\uparrow\rangle + |\downarrow\rangle$$

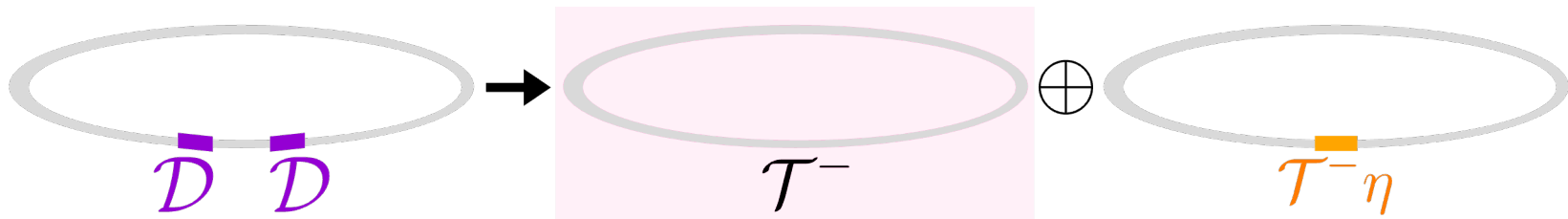


SRE with boundaries $(|f\rangle, |f\rangle)$ is equal to that with $(|\uparrow\rangle + |\downarrow\rangle, |\downarrow\rangle)$

- ◆ reduce EE by a constant (contribution from the boundary entropy)
- ◆ disentangling with Clifford unitaries (CAMPS) might be useless

also for the XXZ chain?

$$\mathcal{D} \otimes \mathcal{D} = \mathcal{T}^- \oplus \mathcal{T}^- \eta$$



SRE with two duality defects is equal to that with a $\mathcal{T}^-$ defect (periodic chain with one less site)

SRE with multiple defects is determined by the fusion rule!

Summary

Nonlinear function of the density matrix (typical in informational quantity)

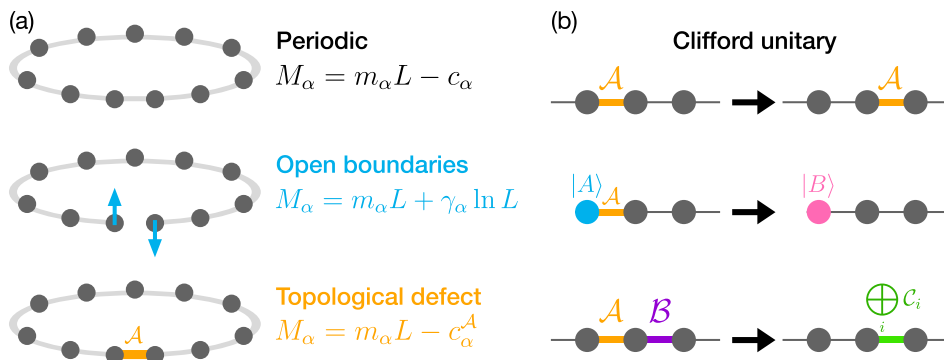
- replica trick & folding trick: express the partition function as amplitudes between boundary states of the multicomponent theory

Stabilizer Rényi Entropy and Conformal Field Theory

	Entanglement	Entanglement (long-range)	Nonstabilizerness	Nonstabilizerness (long-range)
measure	Rényi entropies	mutual information	SRE	mutual SRE
universal behavior	$S_\alpha = \frac{c}{6} \left(1 + \frac{1}{\alpha}\right) \ln l$	$I_\alpha(l) = \frac{c}{3} \left(1 + \frac{1}{\alpha}\right) \ln l$	$M_\alpha(\psi) = m_\alpha L - c_\alpha$	$W_\alpha(l) = \frac{4\Delta_{2\alpha}}{\alpha - 1} \ln l$
characterized by	central charge c	central charge c	g-factor $g_1 = e^{(\alpha-1)c_\alpha}$	scaling dimension of BCCO $\Delta_{2\alpha}$

MH, M. Oshikawa, Y. Ashida, arXiv:2503.13599

SRE with open boundaries and topological defects



SRE reflects universal data of the defects and their fusion rules

MH, Y. Ashida, arXiv:2507.10656

Appendix

Measurements as boundary perturbation

Gibbs state

$$\rho = \frac{e^{-\beta H}}{\text{Tr} e^{-\beta H}}$$

expectation value

$$\langle A \rangle = \text{Tr}[A\rho] = \frac{1}{Z} \int \mathcal{D}\varphi A[\varphi] e^{-S[\varphi]}$$

partition function

$$Z = \int \mathcal{D}\varphi e^{-S[\varphi]}$$

projective measurement



$\{P^{\vec{m}}\}_{\vec{m}}$

post-measurement state

$$\rho^{\vec{m}} = \frac{P^{\vec{m}} \rho P^{\vec{m}}}{p_{\vec{m}}}$$

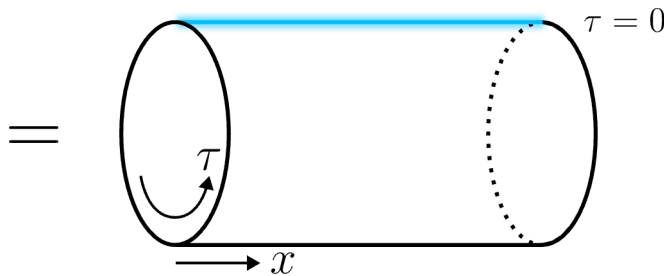
Born probability

$$\begin{aligned} p_{\vec{m}} &= \text{Tr}[P^{\vec{m}} \rho] = \frac{1}{Z} \int \mathcal{D}\varphi P^{\vec{m}}[\varphi] e^{-S[\varphi]} \\ &= \frac{1}{Z} \int \mathcal{D}\varphi e^{-S[\varphi] - \delta S[\varphi]} \end{aligned}$$



$$P^{\vec{m}}[\varphi] = e^{-\delta S[\varphi]}$$

$$Z^{\vec{m}} = \int \mathcal{D}\varphi e^{-S[\varphi] - \delta S[\varphi]}$$



Boundary perturbation $\delta S[\varphi]$

- flows to conformal boundary condition in the IR limit
- infinitely large coupling for projective measurements

S. J. Garratt et al., PRX (2023)

How to determine the boundary condition

participation entropy

$$\sum_{\vec{m}} p_{\vec{m}}^n = \text{Tr} \left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes n} \rho^{\otimes n} \right] = \frac{1}{Z^n} \int \mathcal{D}\vec{\varphi} e^{-\sum_{j=1}^n \mathcal{S}[\varphi_j] - \delta \mathcal{S}_n[\vec{\varphi}]}$$

replicated theory

$$\vec{\varphi} = (\varphi_1, \varphi_2, \dots, \varphi_n)$$

boundary perturbation

$$e^{-\delta \mathcal{S}_n[\vec{\varphi}]} = \sum_{\vec{m}} (P^{\vec{m}})^{\otimes n}$$

Example: Bell-state measurement, $n = 2$

$$\begin{aligned} P^{b_1 b_2} &= |\text{Bell}^{b_1 b_2}\rangle \langle \text{Bell}^{b_1 b_2}| \\ &= \frac{I + (-1)^{b_1} Z Z}{2} \frac{I + (-1)^{b_2} X X}{2} \end{aligned} \quad \longrightarrow \quad \sum_{b_1, b_2} (P^{b_1 b_2})^{\otimes 2} = \frac{I + Z Z^{(1)} Z Z^{(2)}}{2} \frac{I + X X^{(1)} X X^{(2)}}{2}$$

In general,

$$\begin{aligned} \sum_{b_1, b_2} (P^{b_1 b_2})^{\otimes \alpha} &= \prod_{s \in \tilde{S}^{(\alpha)}} \frac{I + s}{2} \\ &\propto \lim_{\mu \rightarrow \infty} e^{\mu \sum s} \end{aligned} \quad \tilde{S}^{(\alpha)} = \{ \underbrace{X X^{(1)} X X^{(2)}}_{\text{interlayer coupling}}, \underbrace{Z Z^{(1)} Z Z^{(2)}}_{\text{interlayer coupling}}, \dots, \underbrace{Z Z^{(\alpha-1)} Z Z^{(\alpha)}}_{\text{interlayer coupling}} \}$$

$$\implies \delta \mathcal{S}_{2\alpha} = -\mu \int d\tau \delta(\tau) \int dx \sum_{s \in \tilde{S}^{(\alpha)}} s$$

should be expressed by operators in the CFT
(requires knowledge of lattice $\leftrightarrow$ CFT correspondence)

Ising CFT: bosonization

Lattice $\leftrightarrow$ CFT $Z \sim \sigma, X - \langle X \rangle \sim \varepsilon$

Bosonization

$$\sigma_1 \sigma_2 = \cos \phi,$$

$$\varepsilon_1 \varepsilon_2 = (\partial_\mu \phi)^2,$$

$$\varepsilon_1 + \varepsilon_2 = \cos 2\phi.$$

$$\mathcal{L} = \frac{1}{2\pi} (\partial_\mu \phi)^2, \quad \phi \sim \phi + 2\pi, \quad \phi \sim -\phi$$

free-boson CFT

compactification $\mathbb{Z}_2$ orbifold



$$\begin{aligned} ZZ^{(i)} ZZ^{(i+1)} &\sim \cos \phi_i \cos \phi_{i+1} \\ &= \cos(\phi_i + \phi_{i+1}) + \cos(\phi_i - \phi_{i+1}) \\ &= 2 \cos(\phi_i - \phi_{i+1}), \end{aligned}$$

$$XX^{(i)} XX^{(i+1)} \sim \langle X \rangle^2 [(\partial_\mu \phi_i)^2 + (\partial_\mu \phi_{i+1})^2]$$

$$\delta \mathcal{S}_{2\alpha} = -\mu \int dx \sum_{i=1}^{\alpha} \cos(\phi_i - \phi_{i+1})$$

**boundary perturbation of α -component CFT
(the most relevant terms)**

Boundary condition (after the folding trick)

$$\Gamma_1 : \begin{cases} \sum_{i=1}^{2\alpha} \phi_i & \text{NBC} \\ \phi_1 - \phi_2 = 0 & \text{DBC} \\ \phi_2 - \phi_3 = 0 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha-1} - \phi_{2\alpha} = 0 & \text{DBC} \end{cases}$$

Neumann of the “center of mass motion”

Dirichlet of the “relative motion”

+ symmetrization for the orbifold

Ising CFT: result

$$\Gamma_2 : \begin{cases} \phi_i - \phi_{i+\alpha} = 0 & \text{DBC} \\ \phi_i + \phi_{i+\alpha} & \text{NBC} \end{cases}$$

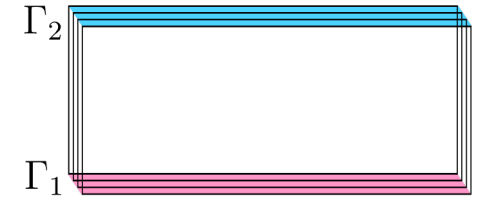
artificial boundary

$$\Gamma_1 : \begin{cases} \sum_{i=1}^{2\alpha} \phi_i & \text{NBC} \\ \phi_1 - \phi_2 = 0 & \text{DBC} \\ \phi_2 - \phi_3 = 0 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha-1} - \phi_{2\alpha} = 0 & \text{DBC} \end{cases}$$

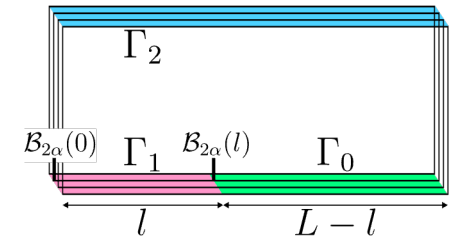
boundary imposed by
Bell-state measurements

$$\Gamma_0 : \begin{cases} \phi_1 = \pi/2 & \text{DBC} \\ \phi_2 = \pi/2 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha} = \pi/2 & \text{DBC} \end{cases}$$

boundary by $|\text{Bell}^{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$
(corresponds to partial trace)



full state SRE



mutual SRE

Constant term of the full state SRE

$$g_1 = \langle \text{GS} | \Gamma_1 \rangle = \sqrt{\alpha} \quad \Rightarrow \quad c_\alpha = \frac{\ln \sqrt{\alpha}}{\alpha - 1}$$

Coefficient of the log-scaling mutual SRE

$$\Delta_{2\alpha} = \frac{1}{16} \quad \Rightarrow \quad W_\alpha(l) = \frac{4\Delta_{2\alpha}}{\alpha - 1} \ln l = \frac{1}{4(\alpha - 1)} \ln l$$

∴ lowest conformal weight included in the amplitude

$$\langle \Gamma_0 | q^{L_0 + \bar{L}_0 - c/12} | \Gamma_1 \rangle = \sum_h n_{\Gamma_0 \Gamma_1}^h \chi_h(\tilde{q}) \quad (n_{\Gamma_0 \Gamma_1}^h \in \mathbb{Z}_{\geq 0})$$

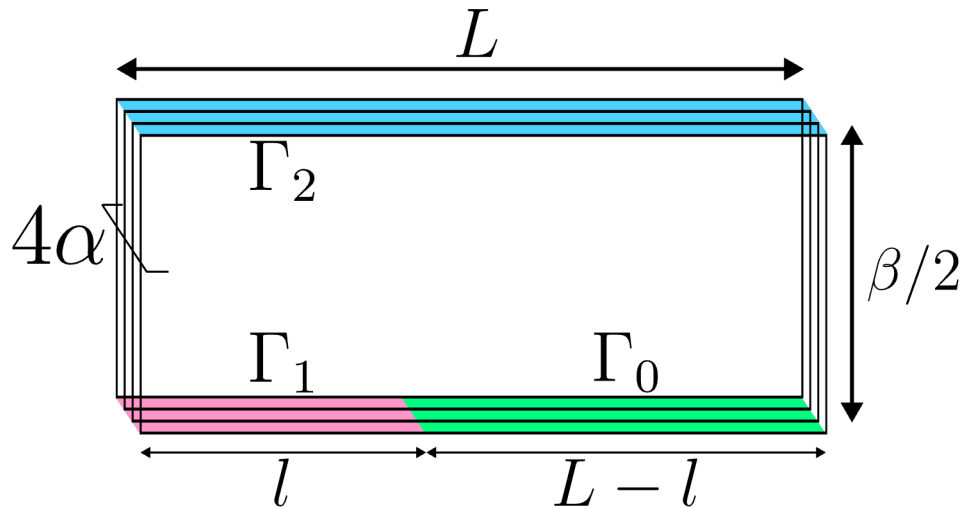
The boundary condition Γ_0

$$\text{Tr}_A[\sigma^{\vec{m}_A} \rho_A] = \text{Tr}[(\sigma^{\vec{m}_A} \otimes I_B) \psi] \quad \vec{m}_A \in \{0, 1\}^{2l}$$

$$\text{Tr}_A^2[\sigma^{\vec{m}_A} \rho_A] = 2^L \text{Tr}[(P^{\vec{m}_A} \otimes (P^{00})^{\otimes L-l})(\psi \otimes \psi^*)]$$

$$M_\alpha(\rho_A) = \frac{(\alpha L - l) \ln 2}{1 - \alpha} + \frac{1}{1 - \alpha} \ln \text{Tr} \left[\sum_{\vec{m}_A} (\tilde{P}^{\vec{m}_A})^{\otimes \alpha} (\psi \otimes \psi^*)^{\otimes \alpha} \right]$$

$$\tilde{P}^{\vec{m}_A} = P^{\vec{m}_A} \otimes (P^{00})^{\otimes (L-l)}$$



Boundary condition imposed by uniform measurement of Bell00

For the Ising model, $|f\rangle \otimes |f\rangle$
 $\parallel$
 $|D(\pi/2)\rangle_{\text{orb}}$

$$\Rightarrow |\Gamma_0\rangle = |D(\pi/2)\rangle_{\text{orb}}^{\otimes 2\alpha}$$

Construction of the boundary states

Ishibashi condition

$$(L_n - \bar{L}_{-n}) |\Gamma\rangle = 0$$

in terms of U(1) currents

$$(\vec{\alpha}_m^L - \mathcal{R} \vec{\alpha}_{-m}^R) |\Gamma\rangle = 0$$

$\mathcal{R} : \mathbf{N} \times \mathbf{N}$ orthogonal matrix

Ishibashi state (coherent state)

$$S(\mathcal{R}) |\vec{R}, \vec{K}\rangle$$

$$S(\mathcal{R}) = \exp \left[- \sum_{n=1}^{\infty} (\vec{a}_n^L)^\dagger \cdot \mathcal{R} (\vec{a}_n^R)^\dagger \right] \quad \vec{K} + \kappa \vec{R} = \mathcal{R} (\vec{K} - \kappa \vec{R})$$

Consistent boundary state of mixed Dirichlet-Neumann boundary condition

$$|\mathcal{R}(\vec{\phi}_D, \vec{\theta}_D)\rangle := g_{\mathcal{R}} \sum_{\vec{R} \in \Lambda_{\mathcal{R}}} \sum_{\vec{K} \in \Lambda_{\mathcal{R}}^*} e^{-i\vec{R} \cdot \vec{\theta}_D - i\vec{K} \cdot \vec{\phi}_D} S(\mathcal{R}) |\vec{R}, \vec{K}\rangle$$

g-factor

Boundary state of the orbifold theory (symmetrization)

$$|\mathcal{R}(\vec{\phi}_D, \vec{\theta}_D)\rangle_{\text{orb}} = \frac{1}{\sqrt{|G|}} \sum_{a \in G} D(a) |\mathcal{R}(\vec{\phi}_D, \vec{\theta}_D)\rangle \quad \text{generic}$$

$$a = \text{diag}(\pm 1, \pm 1, \dots, \pm 1) \in G$$

$$|G| = 2^N$$

$$G_0 = G / \{\pm I\}$$

$$|\mathcal{R}(\vec{\phi}_E, \vec{\theta}_E)\rangle_{\text{orb}} = \sum_{b \in G_0} D(b) \left[\frac{1}{\sqrt{|G|}} |\mathcal{R}(\vec{\phi}_E, \vec{\theta}_E)\rangle \pm 2^{-N/4} |\mathcal{R}(\vec{\phi}_E, \vec{\theta}_E)\rangle_t \right] \quad \text{endpoint}$$